

PKCERT AI & Software Development Internship

Task 05 Report

Data Preprocessing, Visualization & SQL Fundamentals

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Introduction

Data preprocessing is an essential step in data analysis and machine learning. Real-world datasets often contain missing values, duplicate records, inconsistent data, and outliers that can reduce the accuracy of analysis. This project demonstrates a complete workflow for cleaning, visualizing, transforming, and storing data using Python and SQLite. The Titanic dataset was selected because it contains both numerical and categorical features and is widely used for learning data science concepts.

Objective

- Develop practical skills in data preprocessing.
- Create meaningful visualizations.
- Apply feature engineering techniques.
- Perform SQL operations using Python.

Software & Libraries Used

Software: Python 3.x, Visual Studio Code, SQLite.

Libraries: **Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn, sqlite3.**

Dataset Description

Dataset: Titanic Dataset

Source: Kaggle

Description: The dataset contains passenger information including age, gender, passenger class, fare, embarkation port, and survival status. It is commonly used for data science and machine learning practice because it contains missing values and a mix of numerical and categorical features.

```
PS C:\Users\abdul\OneDrive\Desktop\TASK 05> & D:\Python\python.exe
First 5 Rows of the Dataset:
  PassengerId  Survived  Pclass  ...    Fare Cabin Embarked
0           1         0       3  ...   7.2500   NaN        S
1           2         1       1  ...  71.2833   C85        C
2           3         1       3  ...   7.9250   NaN        S
3           4         1       1  ...  53.1000  C123        S
4           5         0       3  ...   8.0500   NaN        S
```

Part A – Data Cleaning & Preprocessing

Step 1: Loading the Dataset

The dataset was loaded into Python using the Pandas library.

Step 2: Checking Missing Values

The dataset contained missing values in several columns, particularly:

- Age
- Cabin
- Embarked

The `isnull()` function was used to identify missing values.

```
Missing Values:
PassengerId      0
Survived         0
Pclass           0
Name             0
Sex              0
Age             177
SibSp            0
Parch           0
Ticket           0
Fare             0
Cabin           687
Embarked         2
dtype: int64
```

Step 3: Handling Missing Values

The following preprocessing techniques were applied:

Age

Missing values in the Age column were replaced with the median value.

Reason:

The median is less affected by extreme values than the mean.

Embarked

Missing values were replaced using the most frequent value (mode).

Reason:

Embarked is a categorical feature.

Cabin

The Cabin column contained a large number of missing values.

Instead of filling them, the entire column was removed.

Reason:

Most values were missing, making the feature less useful.

Step 4: Removing Duplicate Records

The `drop_duplicates()` function was used to remove duplicate rows from the dataset.

Reason:

Duplicate records can produce misleading analysis.

Step 5: Removing Outliers

Outliers in the Age column were detected using the Interquartile Range (IQR) method.

The following formula was used:

- $IQR = Q3 - Q1$
- Lower Limit = $Q1 - 1.5 \times IQR$
- Upper Limit = $Q3 + 1.5 \times IQR$

Values outside this range were removed.

Reason:

Outliers can distort statistical analysis and machine learning models.

Step 6: Saving the Cleaned Dataset

After preprocessing, the cleaned dataset was saved as:

titanic_cleaned.csv

```
Original Dataset Shape: (891, 12)
```

```
Missing Values Before Cleaning:
```

```
PassengerId      0
Survived          0
Pclass           0
Name             0
Sex              0
Age             177
SibSp            0
Parch            0
Ticket           0
Fare             0
Cabin           687
Embarked         2
dtype: int64
```

```
Duplicate Rows: 0
```

```
Missing Values After Cleaning:
```

```
PassengerId      0
Survived          0
Pclass           0
Name             0
Sex              0
Age              0
SibSp            0
Parch            0
Ticket           0
Fare             0
Embarked         0
dtype: int64
```

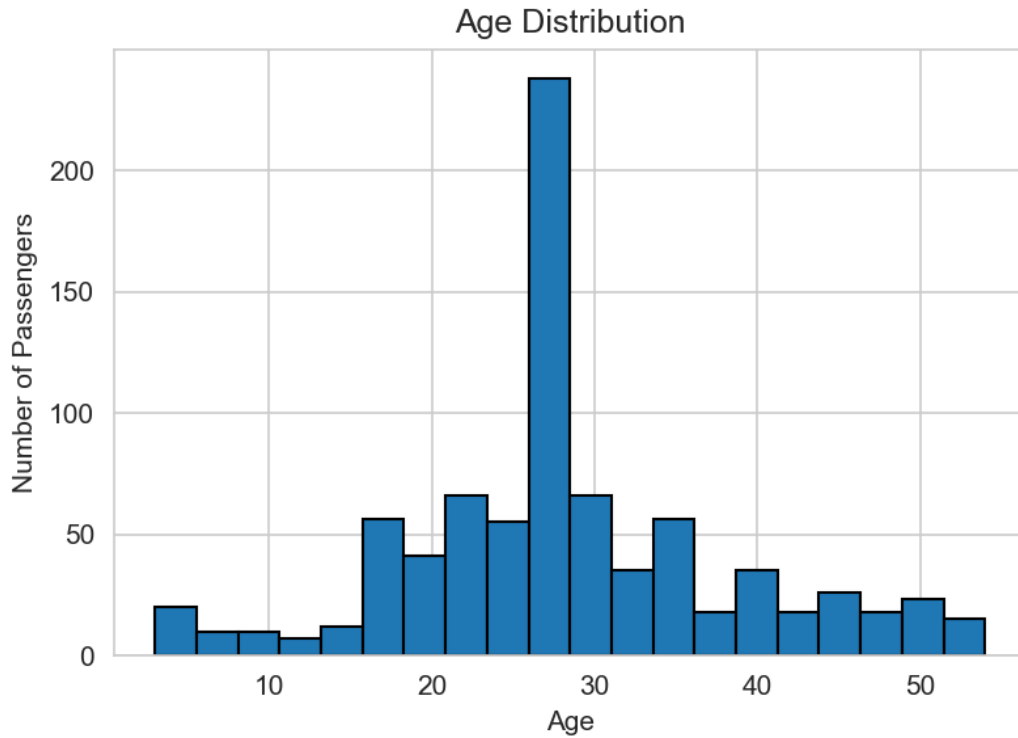
```
Cleaned Dataset Shape: (825, 11)
```

Why these steps?

These preprocessing techniques improve data quality, reduce bias, and prepare the dataset for analysis and machine learning.

Part B – Data Visualization

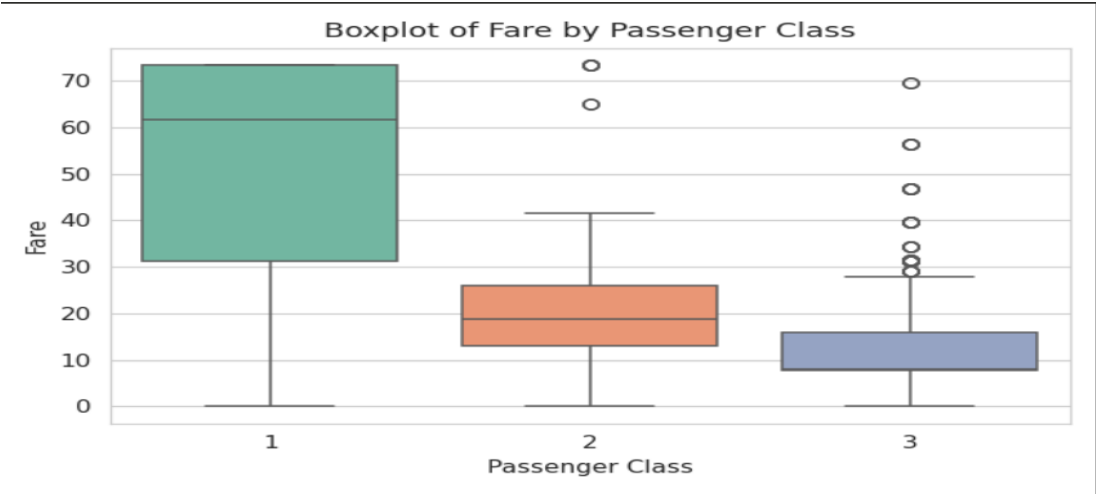
HISTOGRAM:



Histogram: Shows the distribution of passenger ages.

Insight: Most passengers were young adults, while very few passengers were elderly.

BOXPLOT:

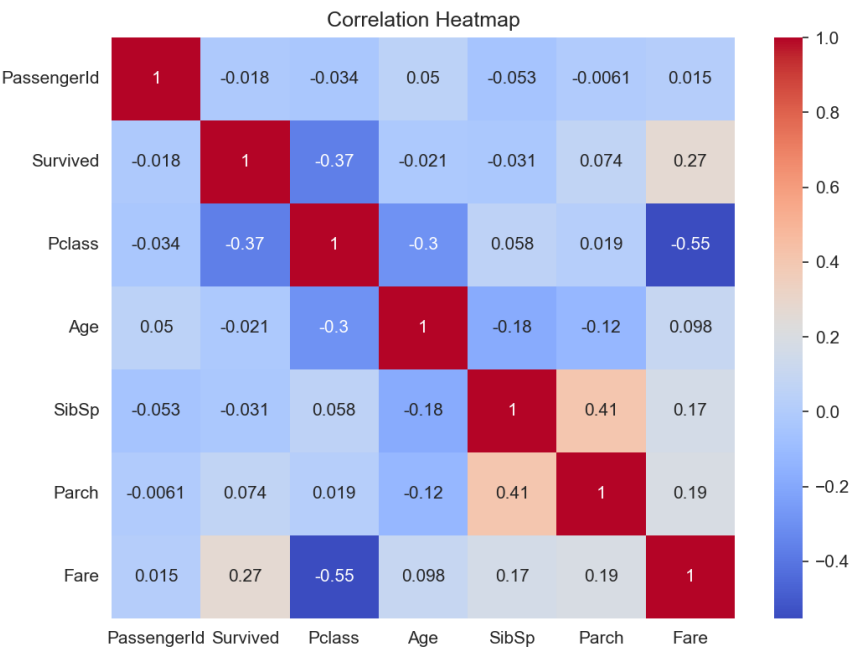


Boxplot: Displays fare distribution and highlights outliers.

Purpose: To detect outliers in ticket fares.

Insight: Several passengers paid significantly higher fares than the majority, indicating the presence of outliers.

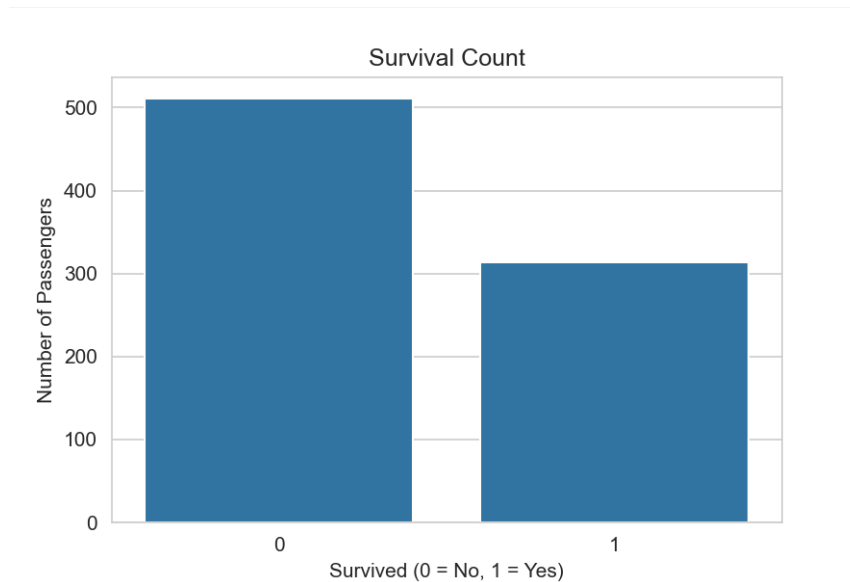
Correlation Heatmap:



Purpose: To understand the relationships between numerical variables.

Insight: The heatmap shows the strength and direction of correlations among numerical features, helping identify related variables.

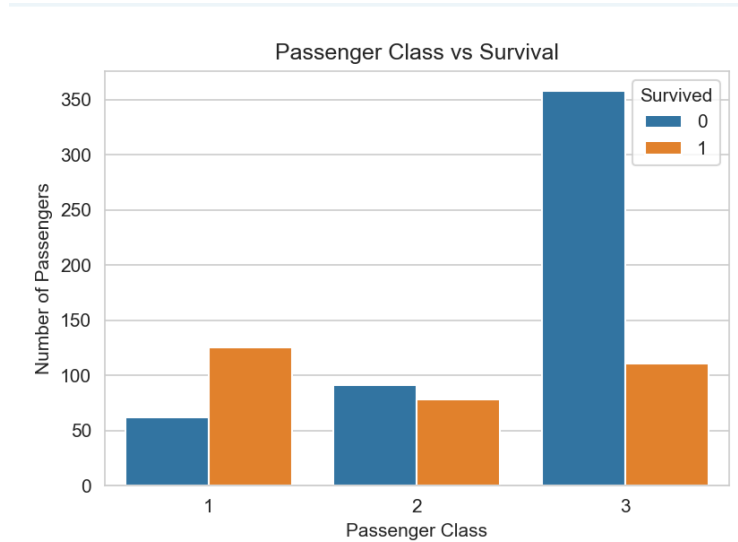
Survival Count Plot: Compares survived vs non-survived passengers.



Purpose: To compare the number of passengers who survived and those who did not.

Insight: The chart shows that more passengers did not survive than survived.

Passenger Class vs Survival: Demonstrates that higher passenger classes generally had better survival rates.



Purpose: To analyze survival rates across passenger classes.

Insight: Passengers traveling in First Class generally had a higher survival rate than those in Third Class.

Part C – Feature Engineering

Categorical features identified: Sex, Embarked, and Pclass.

Applied Label Encoding to Sex and Embarked. Applied One-Hot Encoding to Pclass. Standardized Age and Fare using StandardScaler.

Importance: Encoding converts text values into numerical form required by machine learning algorithms. Standardization scales numerical features to a common range, improving model performance.

Identifying Categorical Features

The categorical features identified in the dataset were:

- Name
- Sex
- Ticket
- Embarked

Among these, **Sex** and **Embarked** were encoded using **Label Encoding**, while **Pclass** was transformed using **One-Hot Encoding**.

Label Encoding

Label Encoding converts text labels into numerical values. It was applied to the **Sex** and **Embarked** columns so they could be processed by machine learning algorithms.

One-Hot Encoding

One-Hot Encoding was applied to the **Pclass** column. It creates separate binary columns for each category, preventing algorithms from assuming an ordinal relationship between passenger classes.

Standardization

The numerical features **Age** and **Fare** were standardized using `StandardScaler`. Standardization transforms features to have a mean of 0 and a standard deviation of 1, ensuring that variables with larger values do not dominate those with smaller values during model training.

OUTPUT:

```
PS C:\Users\abdul\OneDrive\Desktop\TASK 05> & D:\Python\python.exe "c:/Users/abdul/OneDrive/Desktop/TASK 05/partC.py"
Categorical Features:
Index(['Name', 'Sex', 'Ticket', 'Embarked'], dtype='object')

Label Encoding Applied to: Sex, Embarked
One-Hot Encoding Applied to: Pclass

Standardization Applied to: Age and Fare

First 5 Rows After Feature Engineering:
```

File Saved as titanic_Feature Engineernnig

	B	C	D	E	F	G	H	I	J	K	L	M
1	Survived	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Embarked	Pclass_2	Pclass_3	M
2	0	Braund, Mr. Owen Harris	1	-0.64055616	1	0	A/5 21171	-0.48538929	2	FALSE	TRUE	
3	1	Cummings, Mrs. John Bradley (Florence Briggs Thayer)	0	0.932530592	1	0	PC 17599	0.797171219	0	FALSE	FALSE	
4	1	Heikinen, Miss. Laina	0	-0.24728447	0	0	STON/O2. 3101282	-0.47186932	2	FALSE	TRUE	
5	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	0	0.637576826	1	0	113803	0.432967242	2	FALSE	FALSE	
6	0	Allen, Mr. William Henry	1	0.637576826	0	0	373450	-0.46936562	2	FALSE	TRUE	
7	0	Moran, Mr. James	1	-0.05064863	0	0	330877	-0.46118754	1	FALSE	TRUE	
8	0	McCarthy, Mr. Timothy J	1	2.505617341	0	0	17463	0.408180629	2	FALSE	FALSE	
9	1	Johnson, Mrs. Oscar W (Elisabeth Vilhelmina Berg)	0	-0.14896655	0	2	347742	-0.4076084	2	FALSE	TRUE	
10	1	Nasser, Mrs. Nicholas (Adele Achem)	0	-1.42709953	1	0	237736	-0.02829811	0	TRUE	FALSE	
11	1	Sandstrom, Miss. Marguerite Rut	0	-2.41027875	1	1	PP 9549	-0.2961097	2	FALSE	TRUE	
12	0	Saunderscock, Mr. William Henry	1	-0.837192	0	0	A/5. 2151	-0.46936562	2	FALSE	TRUE	
13	0	Andersson, Mr. Anders Johan	1	1.030848513	1	5	347082	-0.00417848	2	FALSE	TRUE	
14	0	Vestrom, Miss. Hulda Amanda Adolfina	0	-1.42709953	0	0	350406	-0.47328742	2	FALSE	TRUE	
15	1	Williams, Mr. Charles Eugene	1	-0.05064863	0	0	244373	-0.37021917	2	TRUE	FALSE	
16	0	Vander Planke, Mrs. Julius (Emelia Maria Vandemoortele)	0	0.244305139	1	0	345763	-0.27007124	2	FALSE	TRUE	
17	1	Masaelmani, Mrs. Fatima	0	-0.05064863	0	0	2649	-0.48589003	0	FALSE	TRUE	
18	0	Fynney, Mr. Joseph J	1	0.637576826	0	0	239865	-0.10983455	2	TRUE	FALSE	
19	1	Beesley, Mr. Lawrence	1	0.539258904	0	0	248698	-0.37021917	2	TRUE	FALSE	
20	1	McGowan, Miss. Anna "Annie"	0	-1.32878161	0	0	330923	-0.46978224	1	FALSE	TRUE	
21	1	Sloper, Mr. William Thompson	1	-0.05064863	0	0	113788	0.080446522	2	FALSE	FALSE	
22	0	Palsson, Miss. Torborg Danira	0	-2.01700706	3	1	349909	-0.20848026	2	FALSE	TRUE	
23	1	Asplund, Mrs. Carl Oscar (Selma Augusta Emilia Johansson)	0	0.932530592	1	5	347077	-0.00192515	2	FALSE	TRUE	
24	0	Emir, Mr. Farred Chehab	1	-0.05064863	0	0	2631	-0.48589003	0	FALSE	TRUE	
25	0	Fortune, Mr. Charles Alexander	1	-0.93550992	3	2	19950	4.637177414	2	FALSE	FALSE	
26	1	O'Dwyer, Miss. Ellen "Nellie"	0	-0.05064863	0	0	330959	-0.47278668	1	FALSE	TRUE	
27	0	Todoroff, Mr. Lalio	1	-0.05064863	0	0	349216	-0.47245418	2	FALSE	TRUE	
28	0	Uruchurtu, Don. Manuel E	1	1.129166435	0	0	PC 17601	-0.07536764	0	FALSE	FALSE	
29	1	Spencer, Mrs. William Augustus (Marie Eugenie)	0	-0.05064863	1	0	PC 17569	2.304147221	0	FALSE	FALSE	
30	1	Gunn, Miss. Mary Azatha	0	-0.05064863	0	0	335677	-0.4753745	1	FALSE	TRUE	

Part D – SQL Fundamentals

Created a SQLite database named titanic.db and imported the cleaned dataset into the passengers table.

Executed the following SQL operations:

- SELECT
- WHERE
- GROUP BY
- JOIN

1. SELECT Query

Purpose: Retrieve the names, ages, and fares of the first five passengers.

```
Database 'titanic.db' created successfully!

===== SELECT QUERY =====
```

	Name	Age	Fare
0	Braund, Mr. Owen Harris	22.0	7.2500
1	Cumings, Mrs. John Bradley (Florence Briggs Th...	38.0	71.2833
2	Heikkinen, Miss. Laina	26.0	7.9250
3	Futrelle, Mrs. Jacques Heath (Lily May Peel)	35.0	53.1000
4	Allen, Mr. William Henry	35.0	8.0500

WHERE Query

Purpose: Retrieve passengers whose fare is greater than 50.

```
===== WHERE QUERY =====
```

	Name	Fare
0	Cumings, Mrs. John Bradley (Florence Briggs Th...	71.2833
1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	53.1000
2	McCarthy, Mr. Timothy J	51.8625
3	Fortune, Mr. Charles Alexander	263.0000
4	Spencer, Mrs. William Augustus (Marie Eugenie)	146.5208
..	...	...
142	Goldenberg, Mrs. Samuel L (Edwiga Grabowska)	89.1042
143	Wick, Mrs. George Dennick (Mary Hitchcock)	164.8667
144	Sage, Miss. Dorothy Edith "Dolly"	69.5500
145	Roebeling, Mr. Washington Augustus II	50.4958
146	Beckwith, Mrs. Richard Leonard (Sallie Monypeny)	52.5542

GROUP BY Query

Purpose: Count passengers based on survival status.

```
===== GROUP BY QUERY =====
```

	Survived	TotalPassengers
0	0	511
1	1	314

JOIN Query

Purpose: Display passenger names along with their passenger class.

```
===== JOIN QUERY =====
```

	Name	ClassName	Survived
0	Braund, Mr. Owen Harris	Third Class	0
1	Cumings, Mrs. John Bradley (Florence Briggs Th...	First Class	1
2	Heikkinen, Miss. Laina	Third Class	1
3	Futrelle, Mrs. Jacques Heath (Lily May Peel)	First Class	1
4	Allen, Mr. William Henry	Third Class	0
5	Moran, Mr. James	Third Class	0
6	McCarthy, Mr. Timothy J	First Class	0
7	Johnson, Mrs. Oscar W (Elisabeth Vilhelmina Berg)	Third Class	1
8	Nasser, Mrs. Nicholas (Adele Achem)	Second Class	1
9	Sandstrom, Miss. Marguerite Rut	Third Class	1

Results

The dataset was successfully cleaned by handling missing values, removing duplicates, and eliminating outliers. Visualizations revealed patterns in passenger ages, fares, survival counts, and class-wise survival. Feature engineering converted categorical variables into numerical form and standardized numerical features. The cleaned dataset was stored in SQLite, and SQL queries were executed successfully through Python.

Conclusion

This project successfully demonstrated data preprocessing, visualization, feature engineering, and SQL integration using Python. The cleaned dataset is suitable for future analysis and machine learning tasks. All objectives of the task were achieved successfully.